Họ và tên: Lương Cao Pháp - 24521288

ASSIGNMENT 4

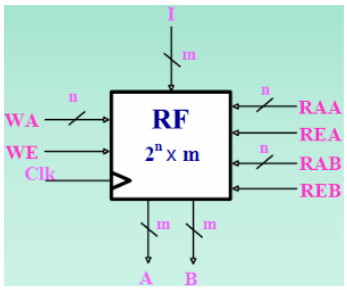
Vẽ mạch thiết kế bộ nhớ: Register file 4x4 (1 write port, 2 read port) và RAM 4x4 như thiết kế trình bày trong bài học. Vẽ mạch và mô phỏng kết quả đọc ghi bộ nhớ trên Quartus.

7.16 (English textbook)

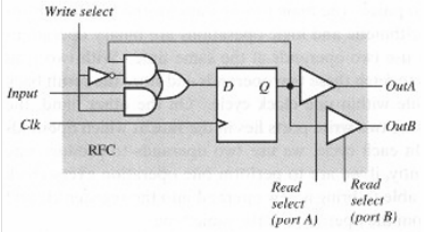
Thiết kế Register file 4x4 (1 write port, 2 read port)

Register-file with 1 write port and 2 read ports

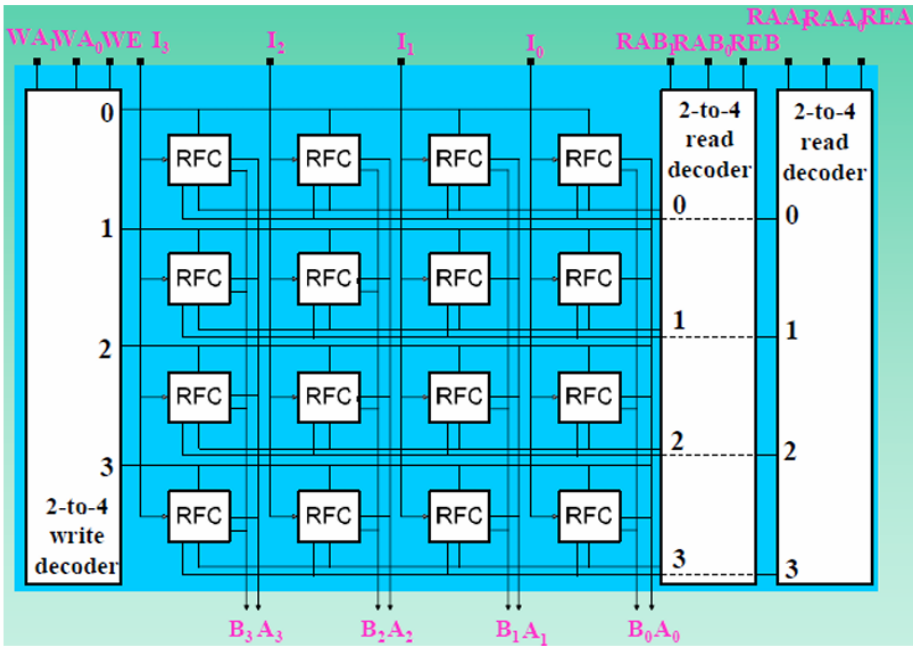
- This register-file is used for reading two operands and writing one result in each clock cycle.



Graphic symbol

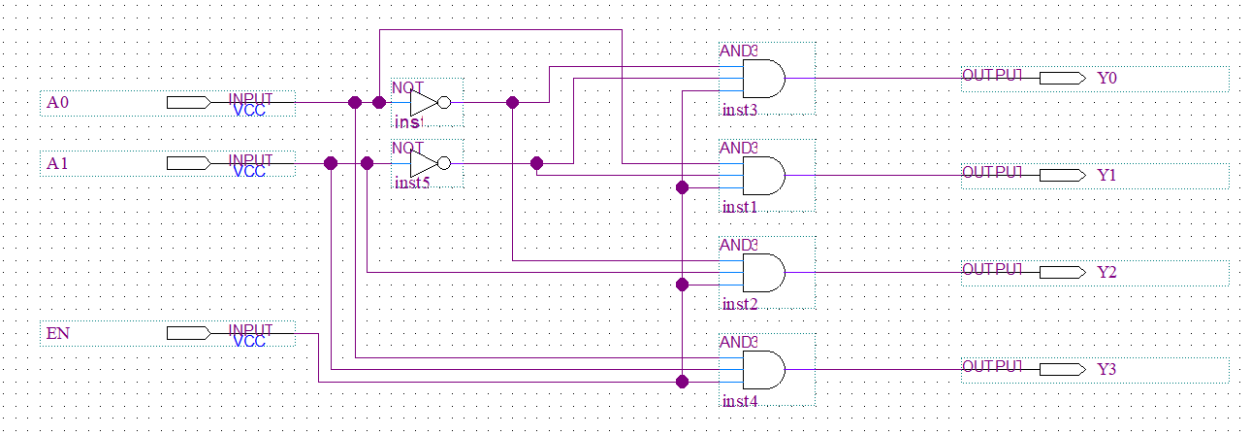


Register-file cell

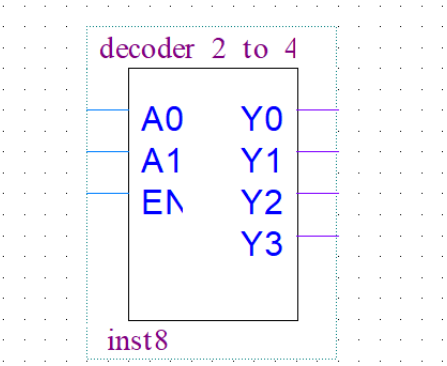


Logic schematic

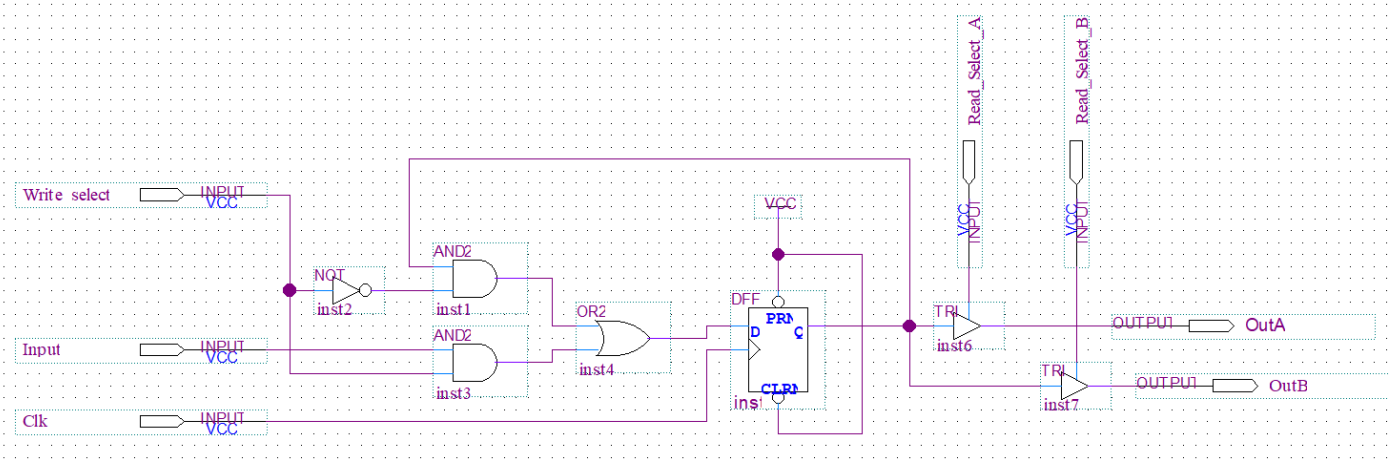
a. Thiết kế và vẽ mạch trên Quartus



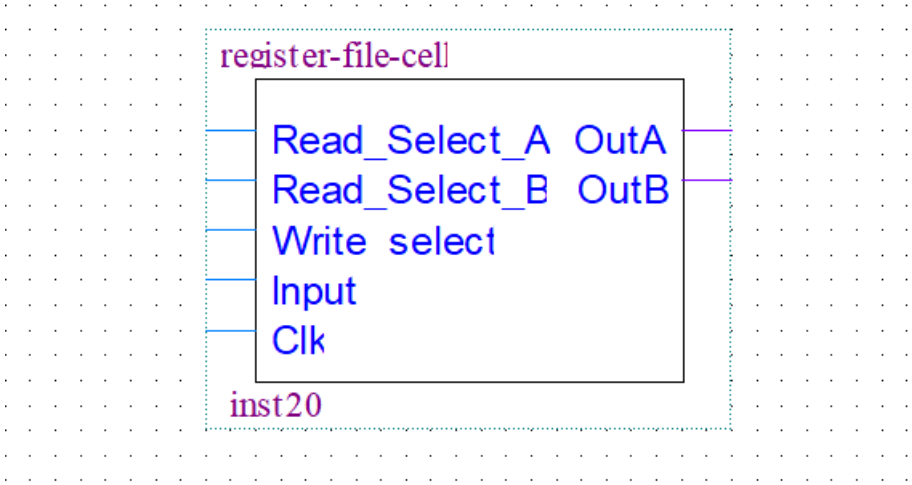
Hình 1. Vẽ mạch Decoder 2 to 4 có Enable



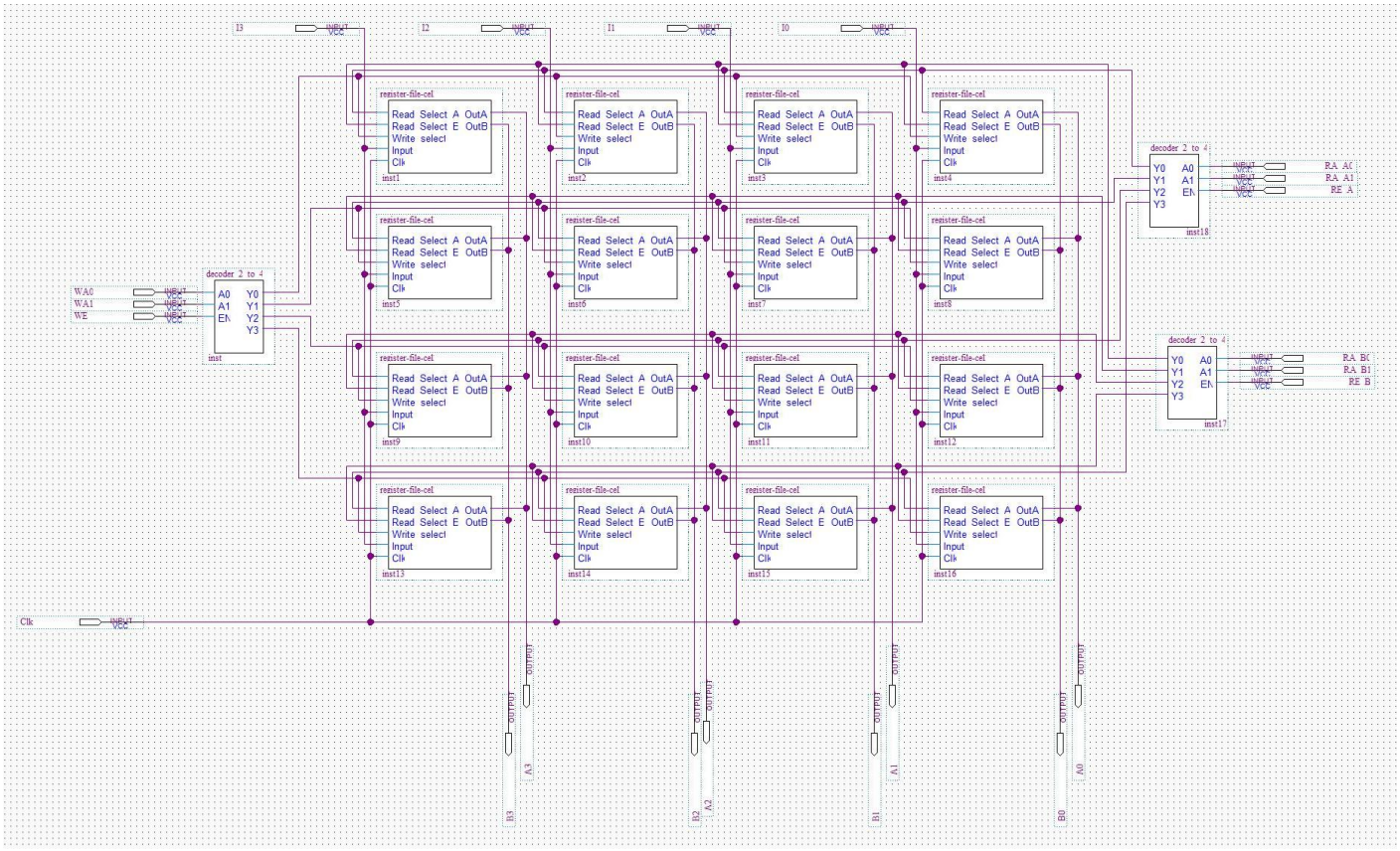
Hình 2. Đóng gói mạch Decoder 2 to 4 có Enable



Hình 3. Vẽ mạch Register File Cell

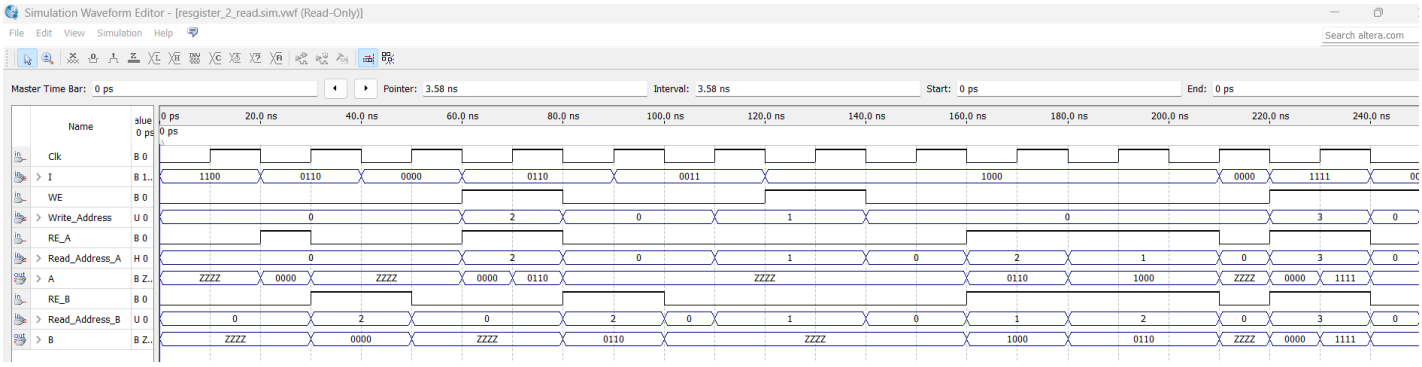


Hình 4. Đóng gói mạch Register File Cell



Hình 5. Kết hợp và vẽ mạch Register file 4x4 (1 write port, 2 read port)

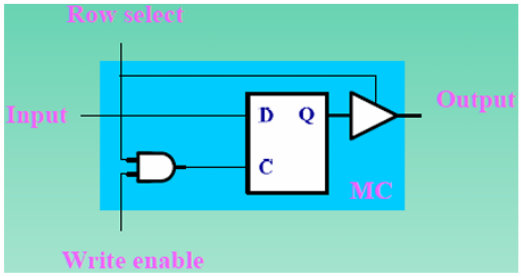
b. Mô phỏng Register file 4x4 (1 write port, 2 read port)



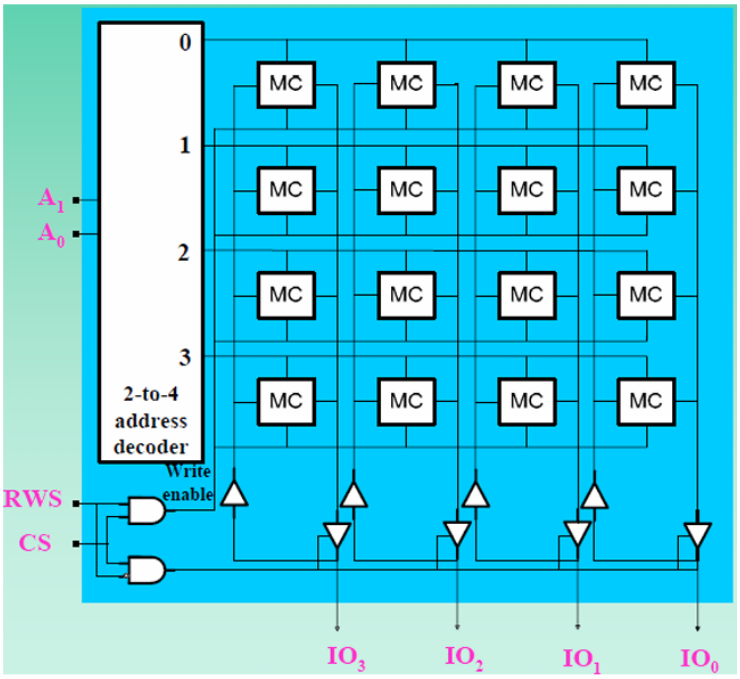
Thiết kế RAM 4x4

RAM organization

- Ram memory cells can be static or dynamic.
- Static RAM do not lose data with time.
- Dynamic RAM must be refreshed.

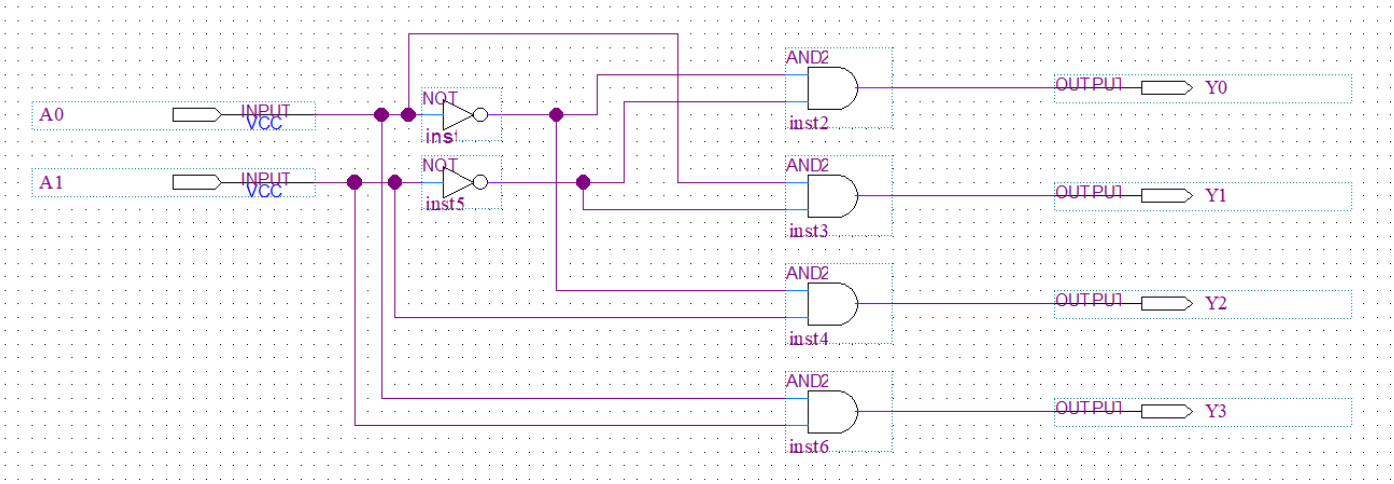


Memory cell

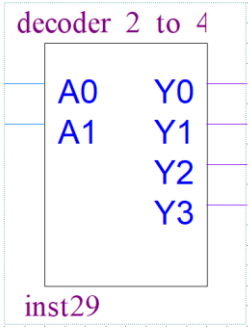


Memory schematic

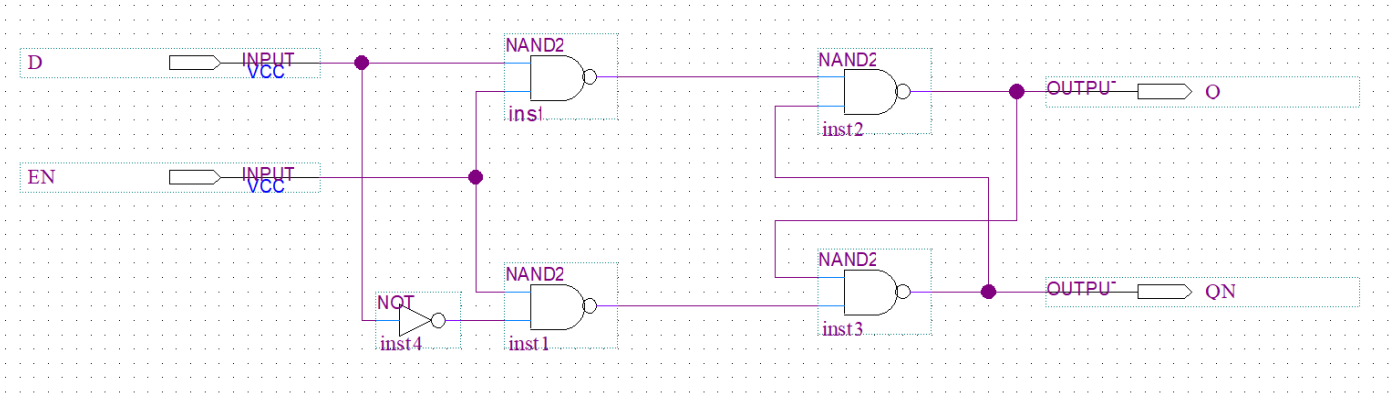
a. Thiết kế và vẽ mạch trên Quartus



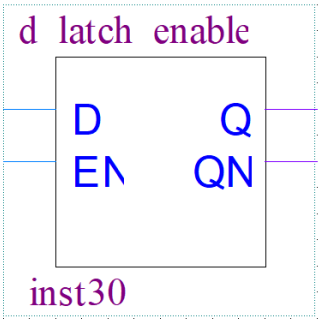
Hình 6. Thiết kế mạch Decoder 2 to 4 (không có Enable)



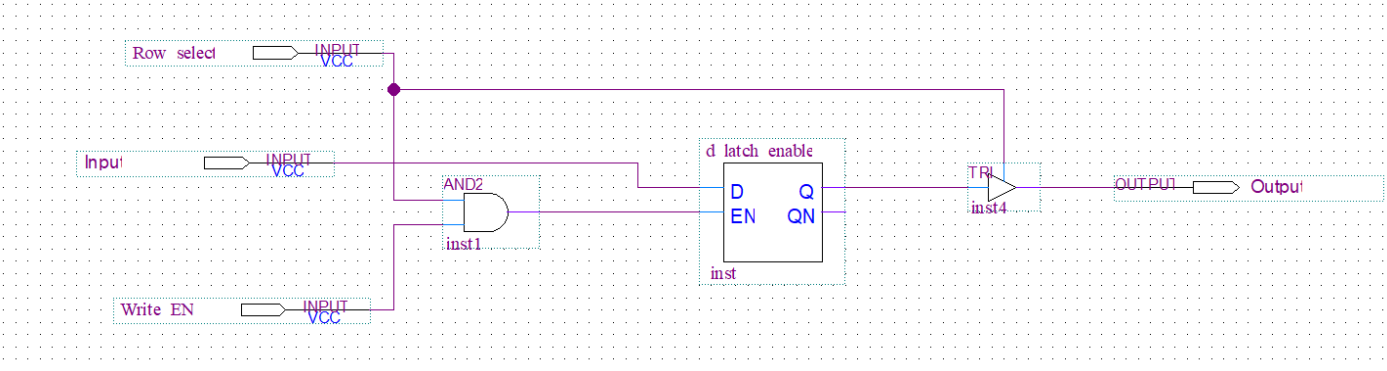
Hình 7. Đóng gói mạch Decoder 2 to 4 (không có Enable)



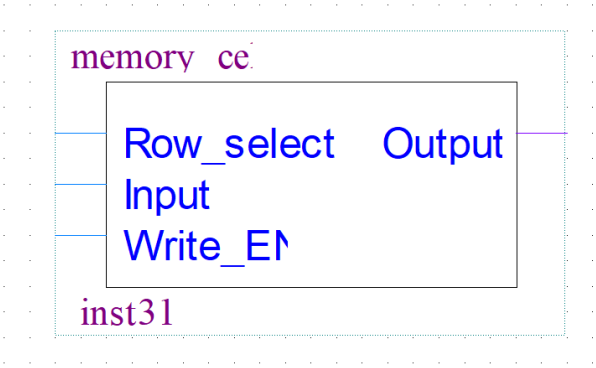
Hình 8. Thiết kế D Latch có Enable



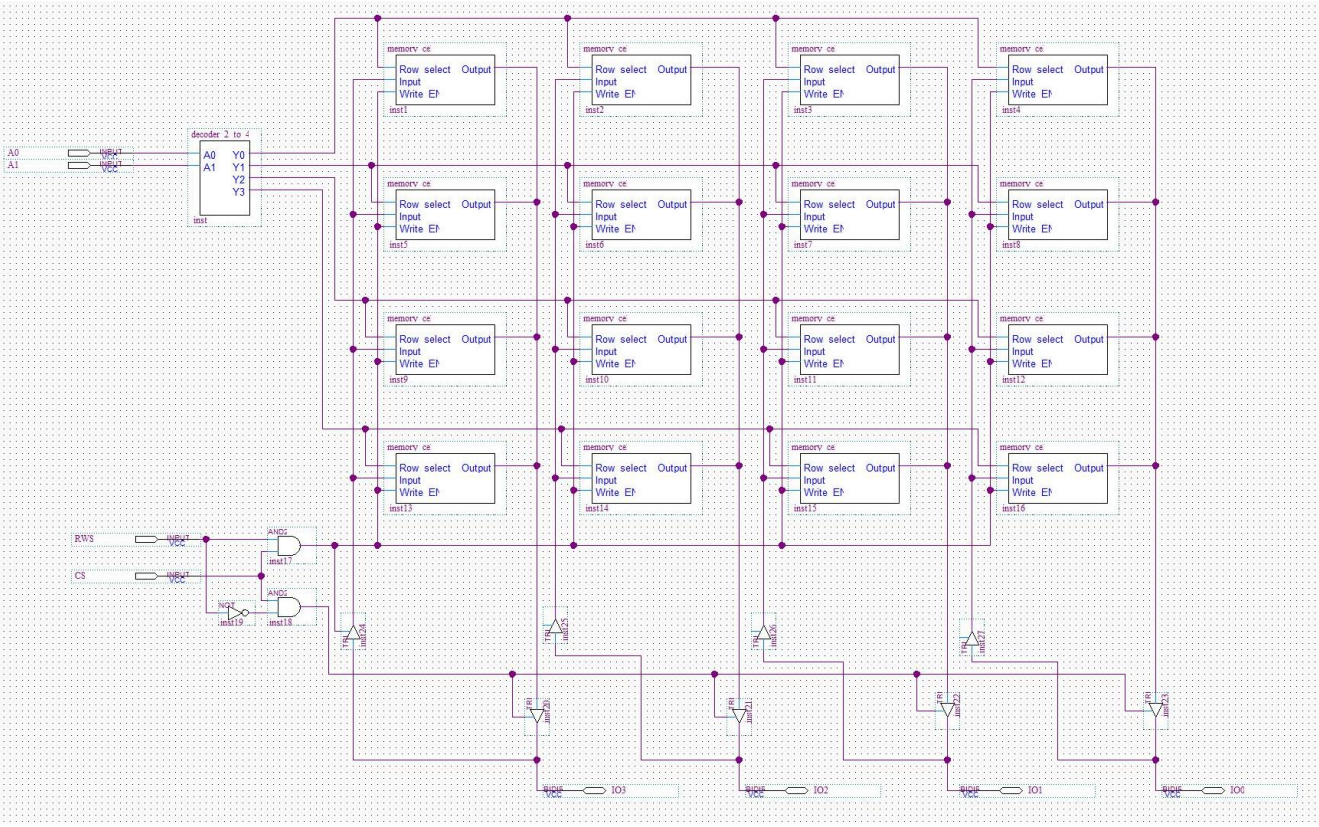
Hình 9. Đóng gói D Latch có Enable



Hình 10. Thiết kế mạch Memory Cell

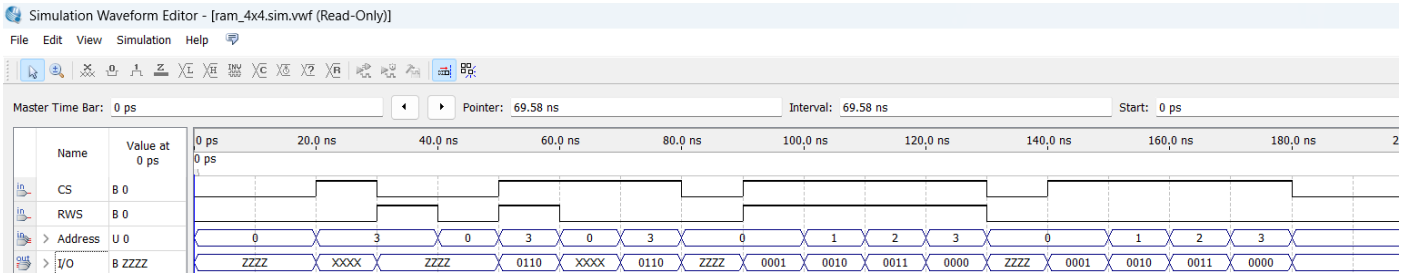


Hình 11. Đóng gói mạch Memory Cell



Hình 12. Vẽ mạch RAM 4x4

b. Mô phỏng RAM 4x4



7.16 (Memories) Design:

(a) 256K × 8 RAM using 256K × 1 RAM chips

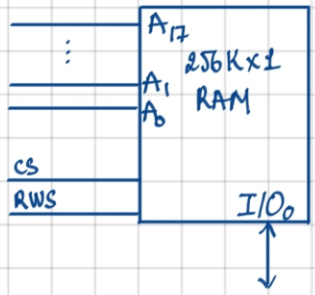
(b) 64K × 32 RAM using 64K × 8 RAM chips

(c) 1M × 1 RAM using 256K × 1 RAM chips

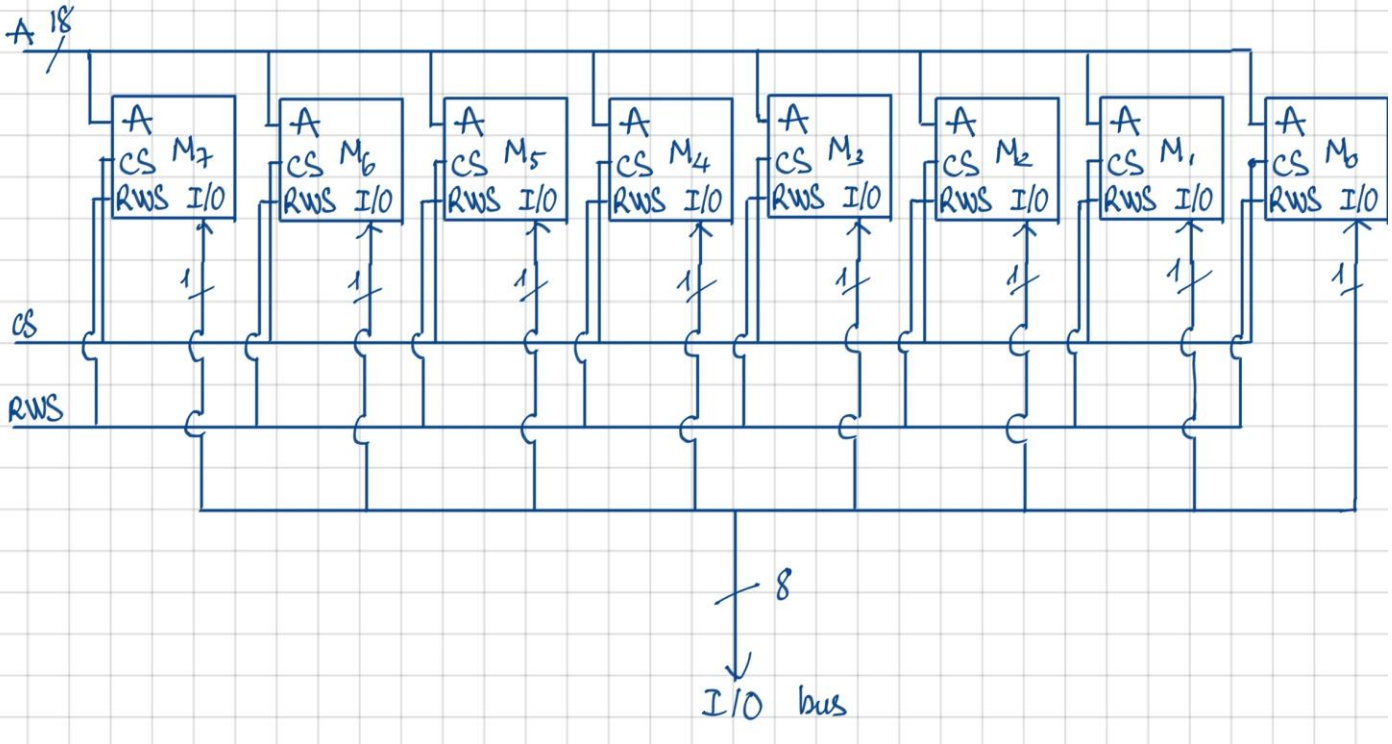
(d) 256K × 8 RAM using 64K × 8 RAM chips

a) Design 256K x 8 RAM using 256K x 1 RAM chips

* 256K x 1 RAM

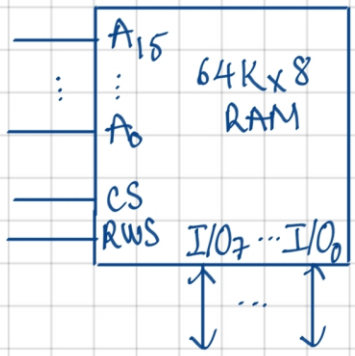


* 256K x 8 RAM

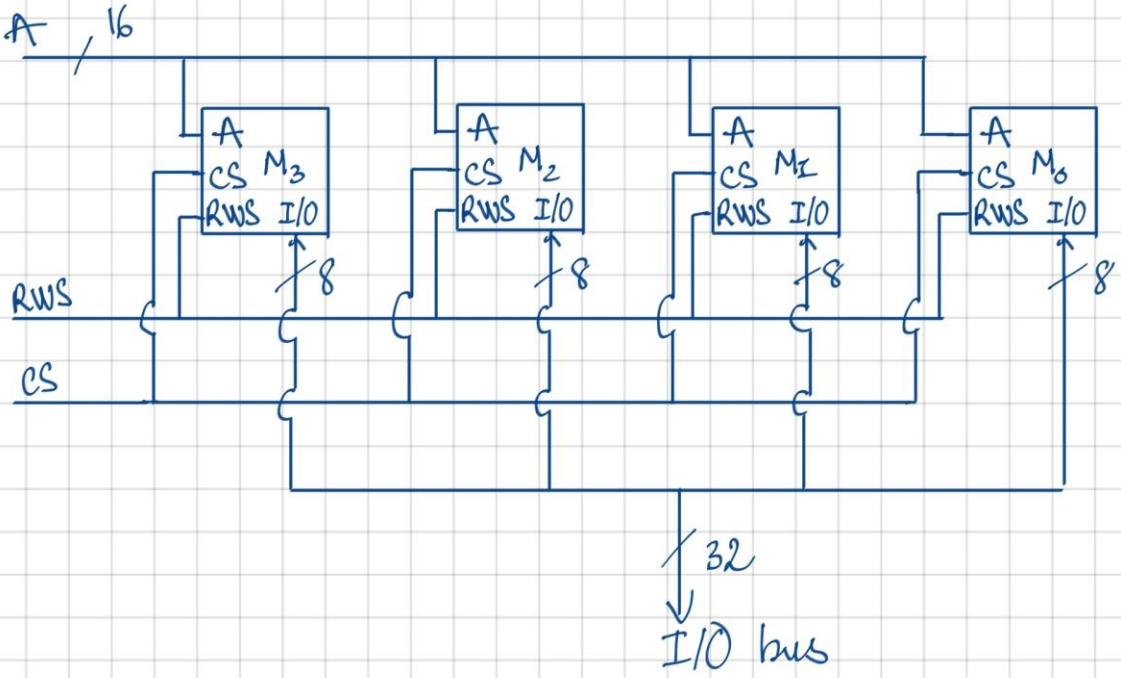


6) 64K x 32 RAM using 64K x 8 RAM chips

* 64K x 8 RAM chips

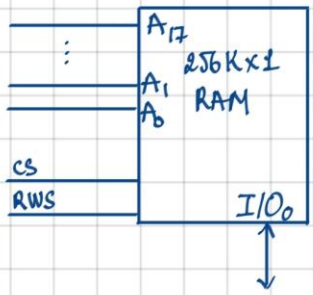


* 64K x 32 RAM

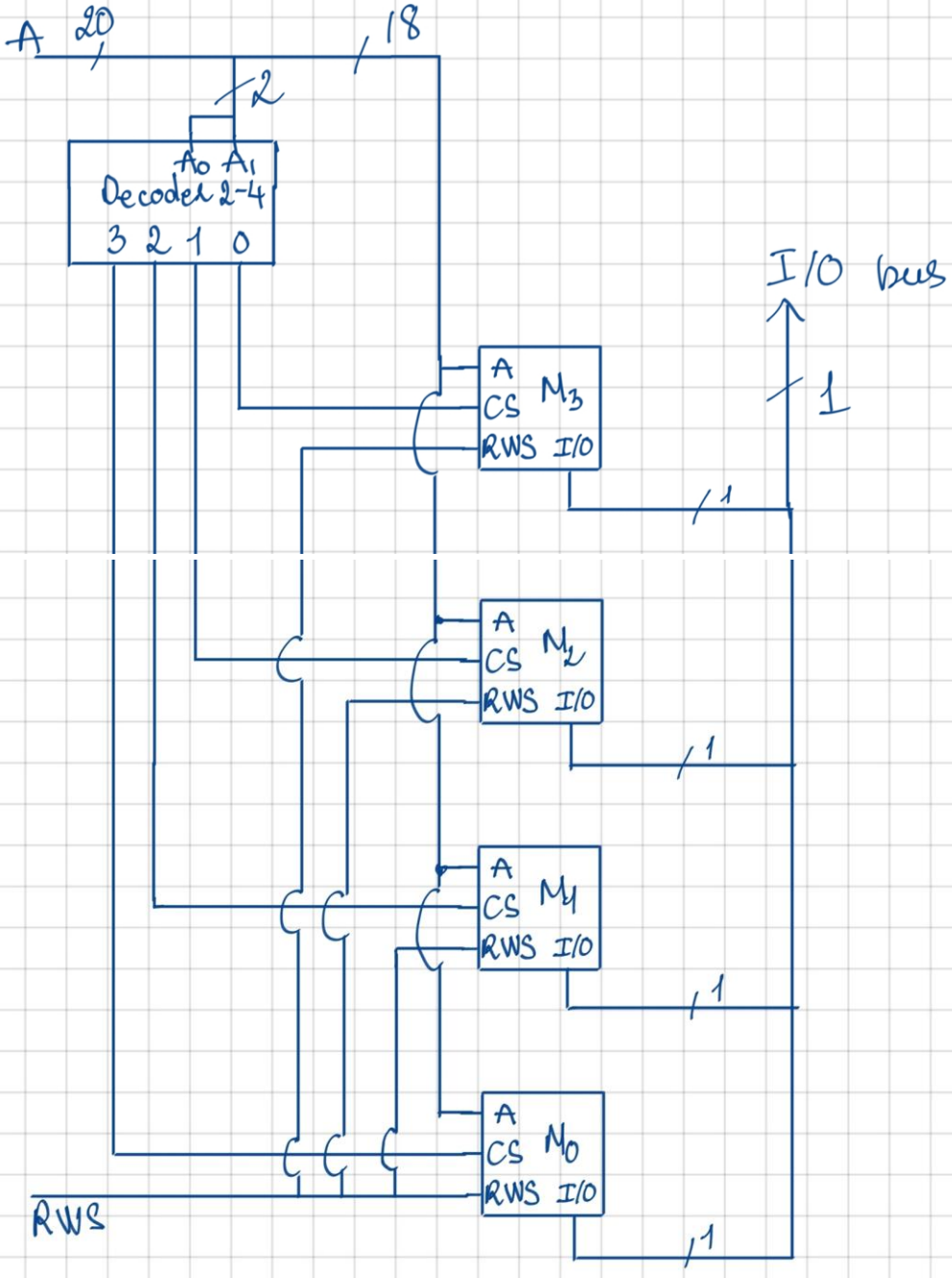


c) 1M x 1 RAM using 256K x 1 RAM chips

* 256K x 1 RAM chips

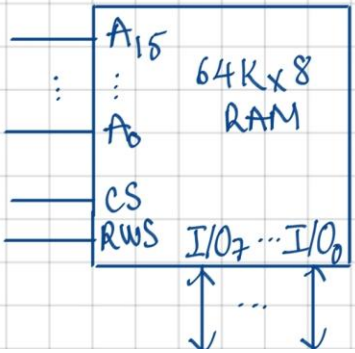


* 1M x 1 RAM



d) 256K x 8 RAM using 64K x 8 RAM clips

* 64K x 8 RAM chips



* 256K x 8RAM using 64K x 8RAM chips

